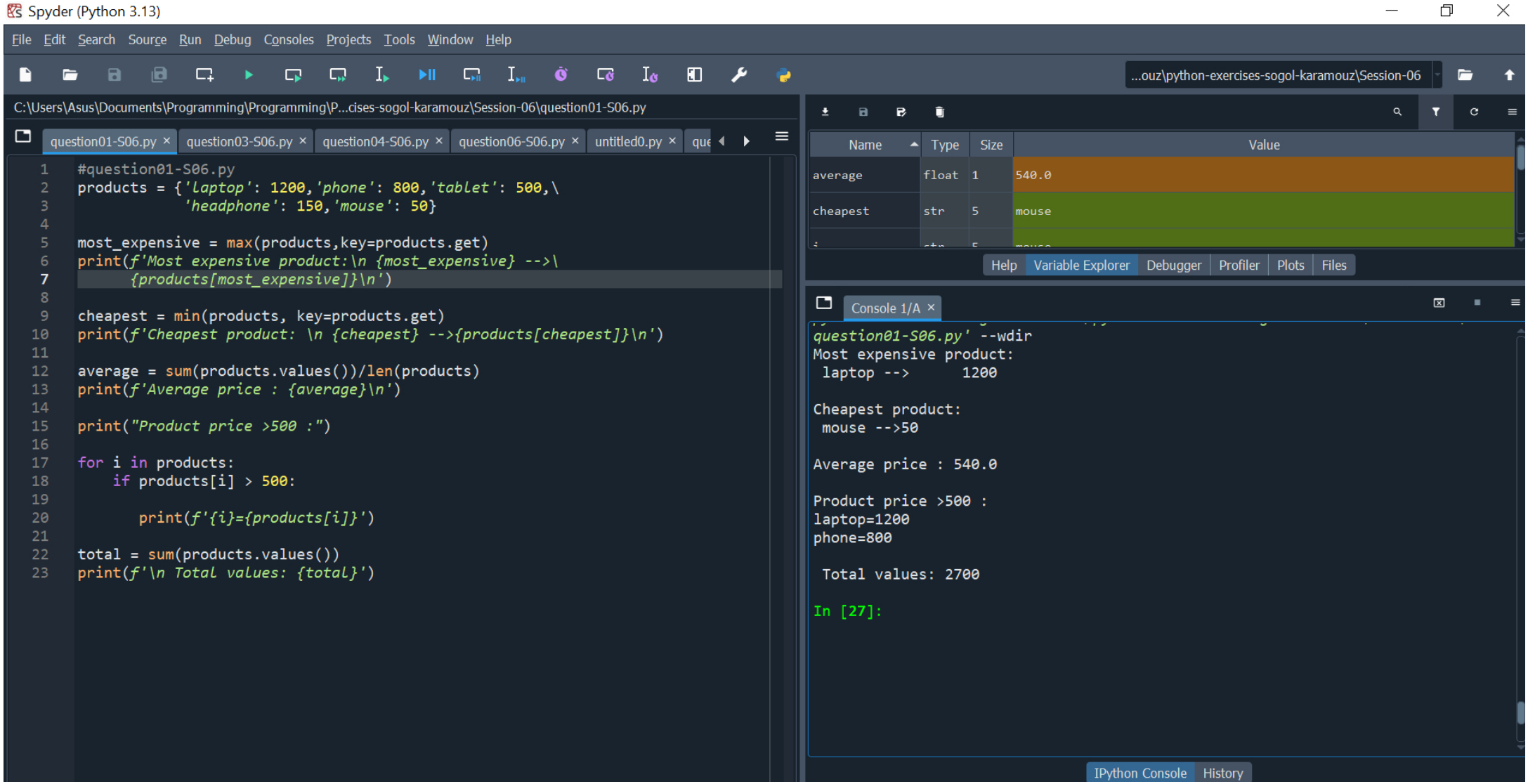
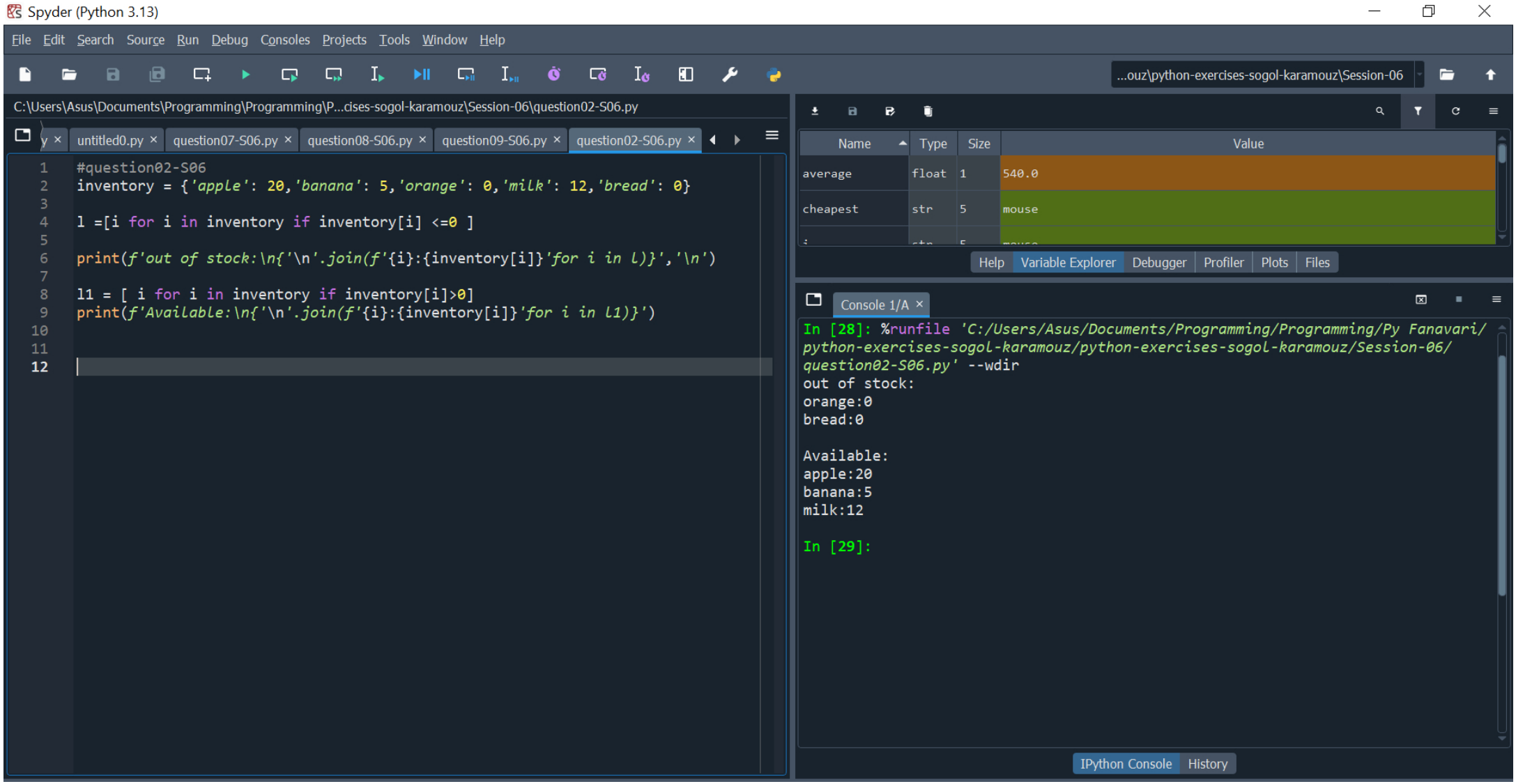
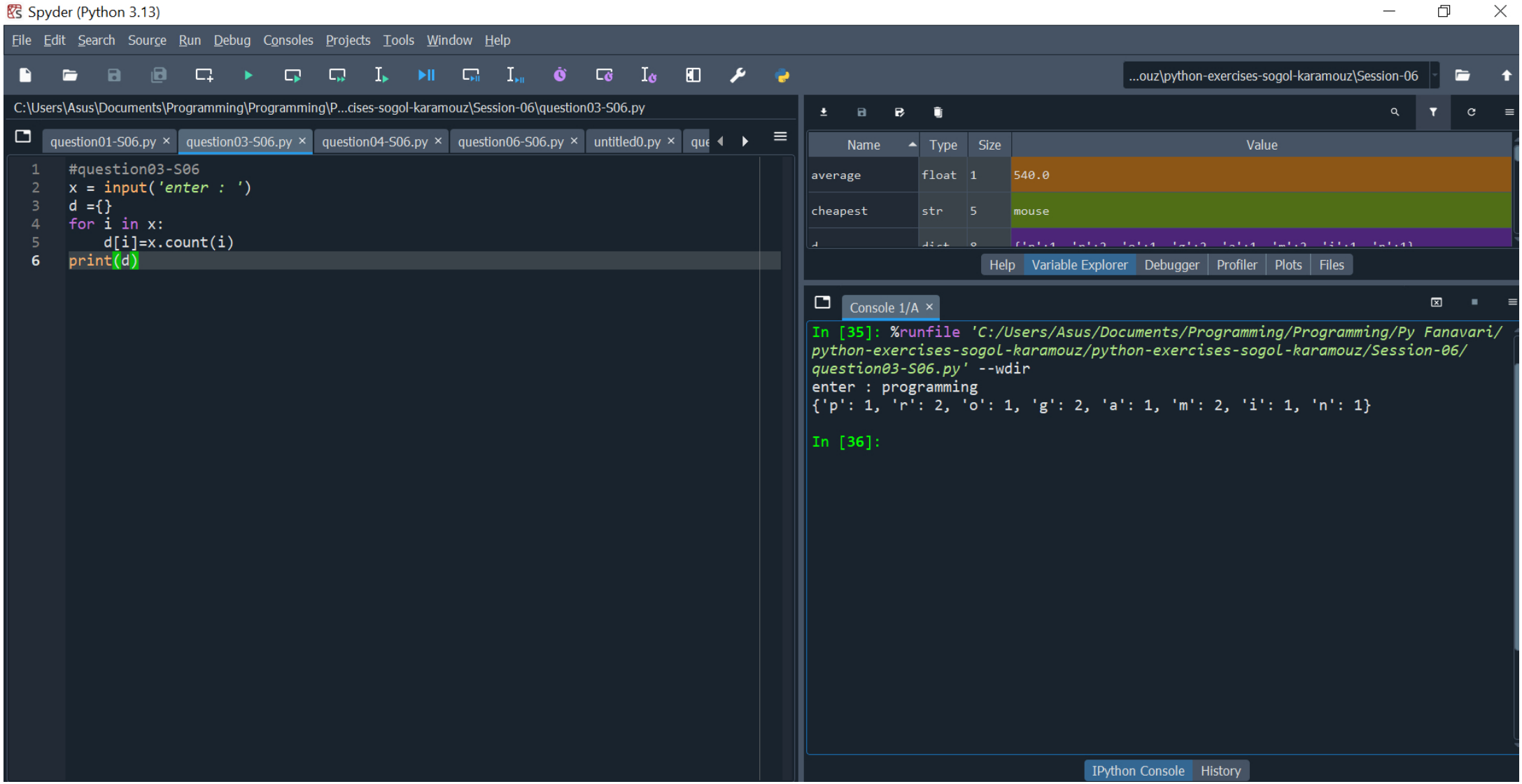
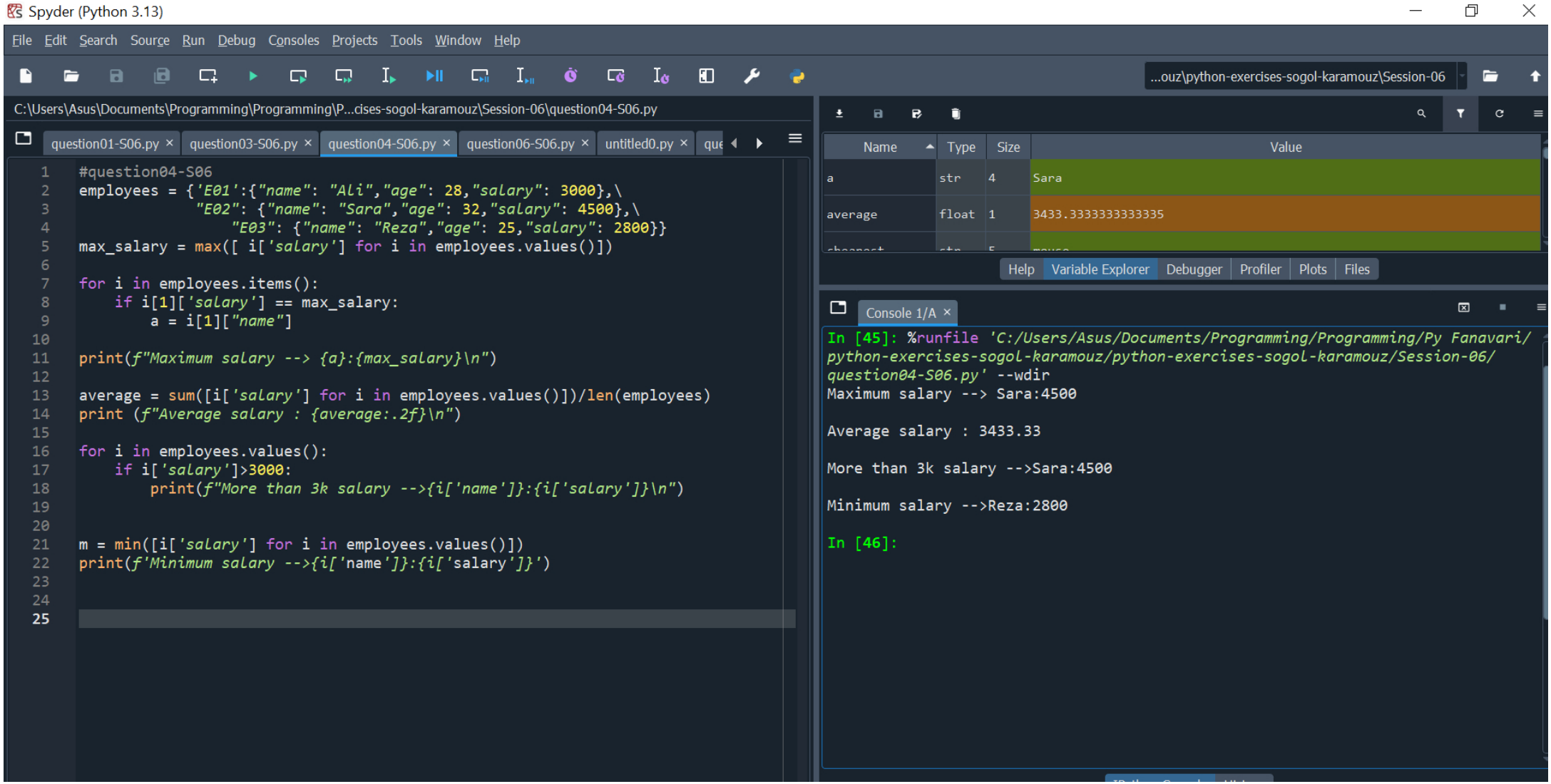
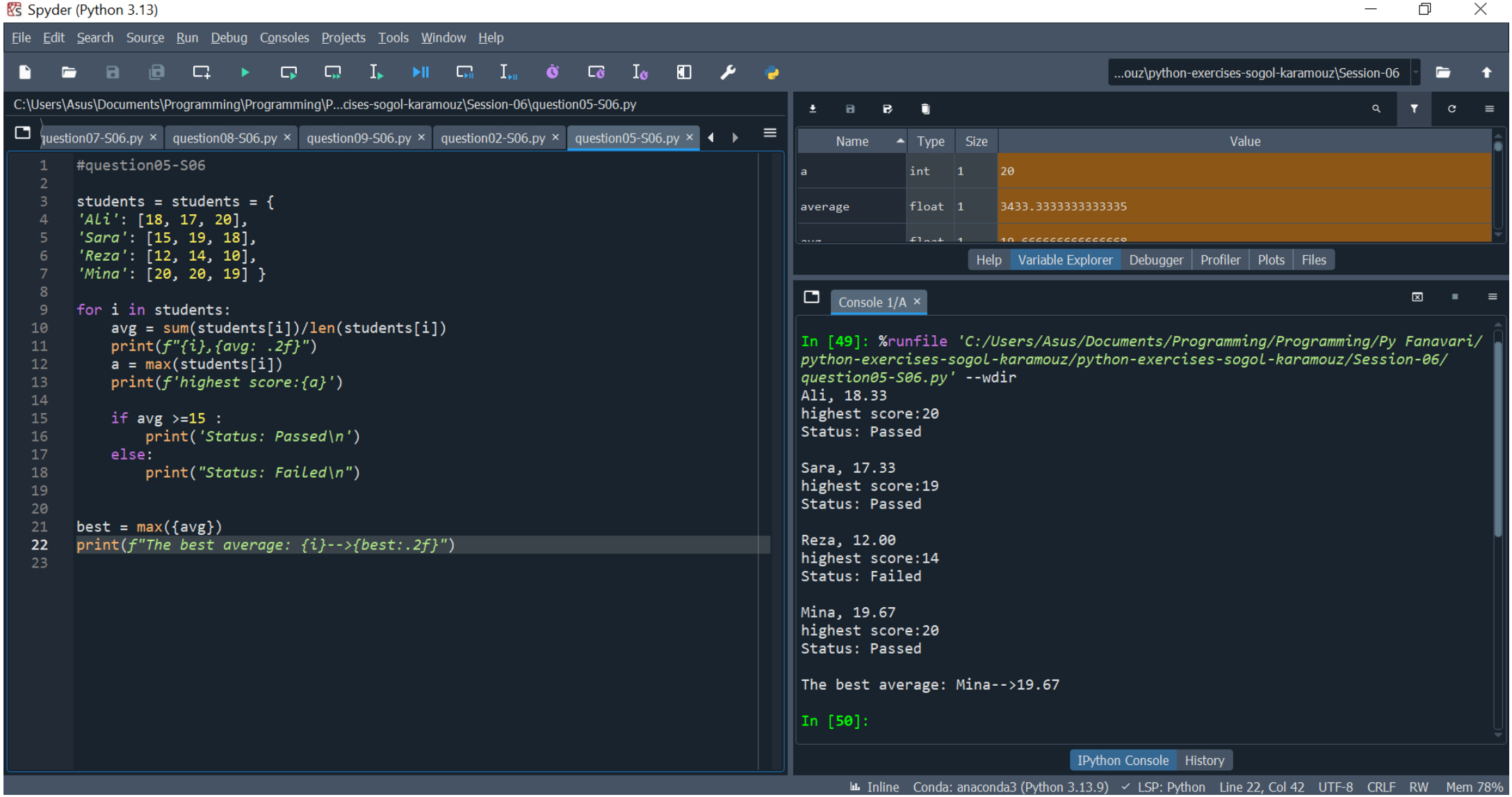
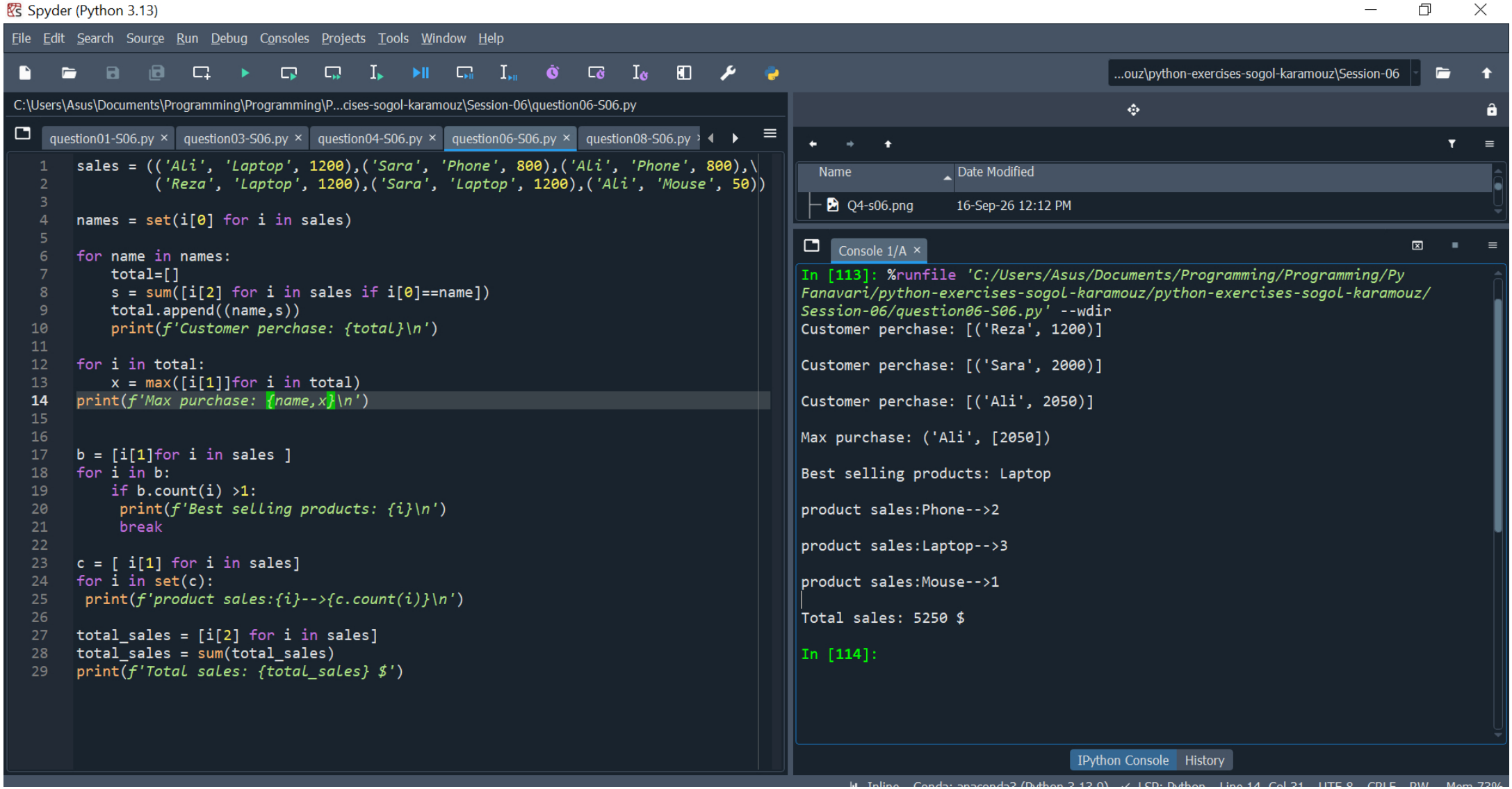

Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question01-S06.py
question01-S06.py × question03-S06.py × question04-S06.py × question06-S06.py × untitled0.py × que
#question01-S06.py
products = {'laptop': 1200,'phone': 800,'tablet': 500,\
'headphone': 150,'mouse': 50}
most_expensive = max(products,key=products.get)
print(f'Most expensive product:\n {most_expensive} -->\
{products[most_expensive]}\n')
cheapest = min(products, key=products.get)
print(f'Cheapest product: \n {cheapest} -->{products[cheapest]}\n')
average = sum(products.values())/len(products)
print(f'Average price : {average}\n')
print("Product price >500 :")
for i in products:
if products[i] > 500:
print(f'{i}={products[i]}')
total = sum(products.values())
print(f'\n Total values: {total}')
Name Type Size Value
average float 1 540.0
cheapest str 5 mouse
Help Variable Explorer Debugger Profiler Plots Files
Console 1/A ×
question01-S06.py' --wdir
Most expensive product:
laptop --> 1200
Cheapest product:
mouse -->50
Average price : 540.0
Product price >500 :
laptop=1200
phone=800
Total values: 2700
In [27]:
IPython Console History

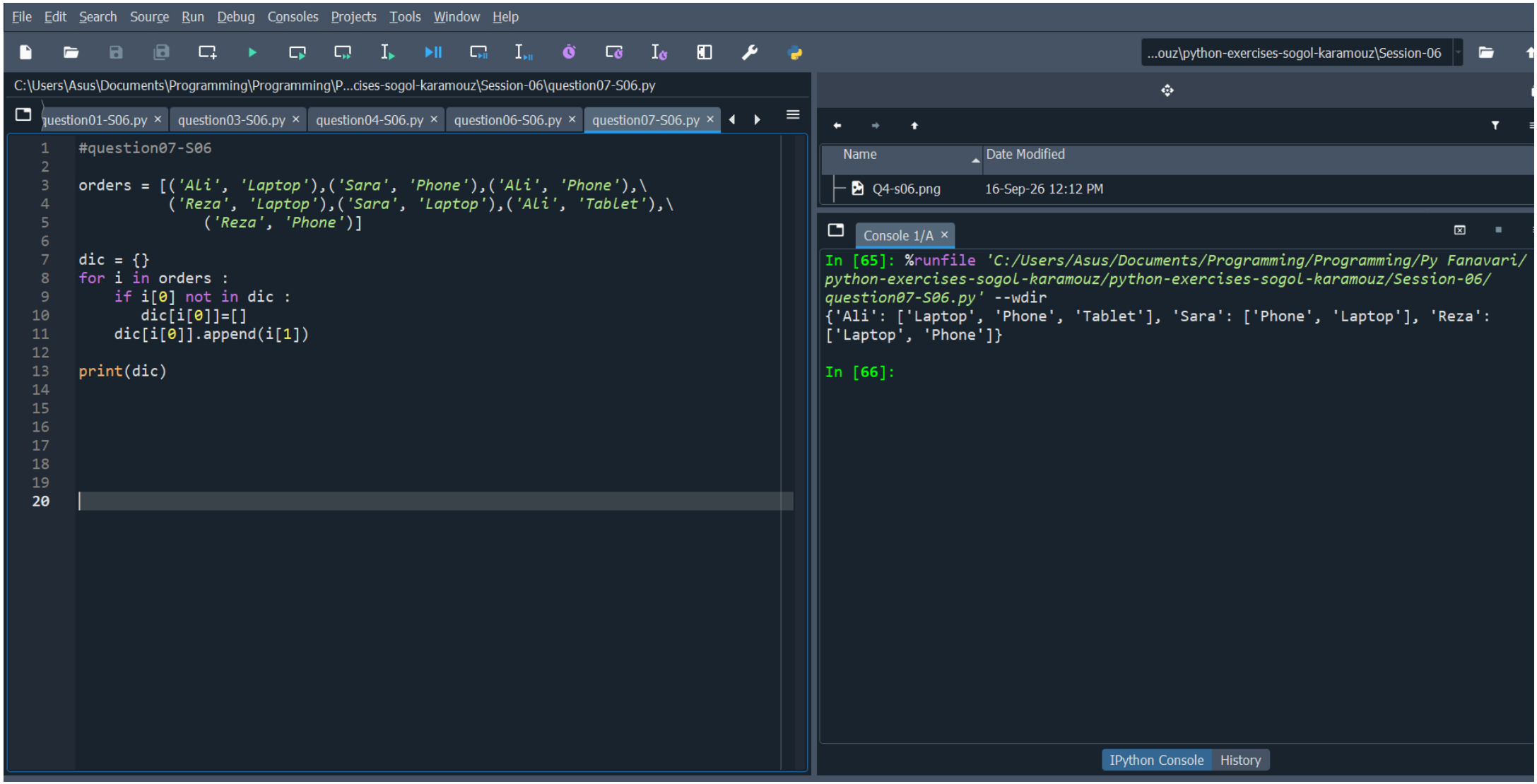

Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question02-S06.py
untitled0.py
question07-S06.py
question08-S06.py
question09-S06.py
question02-S06.py
#question02-S06
inventory = {'apple': 20,'banana': 5,'orange': 0,'milk': 12,'bread': 0}
l =[i for i in inventory if inventory[i] <=0]
print(f'out of stock:\n{'\n'.join(f'{i}:{inventory[i]}'for i in l)}','\n')
l1 = [i for i in inventory if inventory[i]>0]
print(f'Available:\n{'\n'.join(f'{i}:{inventory[i]}'for i in l1)}')
Name
Type
Size
Value
average
float
1
540.0
cheapest
str
5
mouse
Help
Variable Explorer
Debugger
Profiler
Plots
Files
Console 1/A
In [28]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/Session-06/question02-S06.py' --wdir
out of stock:
orange:0
bread:0
Available:
apple:20
banana:5
milk:12
In [29]:
IPython Console
History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question03-S06.py
question01-S06.py
question03-S06.py
question04-S06.py
question06-S06.py
untitled0.py
#question03-S06
x = input('enter : ')
d ={}
for i in x:
d[i]=x.count(i)
print(d)
Name Type Size Value
average float 1 540.0
cheapest str 5 mouse
Help Variable Explorer Debugger Profiler Plots Files
Console 1/A
In [35]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/Session-06/question03-S06.py' --wdir
enter : programming
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
In [36]:
IPython Console History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question04-S06.py
question01-S06.py
question03-S06.py
question04-S06.py
question06-S06.py
untitled0.py
#question04-S06
employees = {'E01':{"name": "Ali","age": 28,"salary": 3000},\
"E02": {"name": "Sara","age": 32,"salary": 4500},\
"E03": {"name": "Reza","age": 25,"salary": 2800}}
max_salary = max([i['salary'] for i in employees.values()])
for i in employees.items():
if i[1]['salary'] == max_salary:
a = i[1]["name"]
print(f"Maximum salary --> {a}:{max_salary}\n")
average = sum([i['salary'] for i in employees.values()])/len(employees)
print (f"Average salary : {average:.2f}\n")
for i in employees.values():
if i['salary']>3000:
print(f"More than 3k salary -->{i['name']}:{i['salary']}\n")
m = min([i['salary'] for i in employees.values()])
print(f'Minimum salary -->{i['name']}:{i['salary']}')
Name
Type
Size
Value
a
str
4
Sara
average
float
1
3433.3333333333335
Help
Variable Explorer
Debugger
Profiler
Plots
Files
Console 1/A
In [45]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/Session-06/question04-S06.py' --wdir
Maximum salary --> Sara:4500
Average salary : 3433.33
More than 3k salary -->Sara:4500
Minimum salary -->Reza:2800
In [46]:


Spyder (Python 3.13)
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...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question05-S06.py
question07-S06.py
question08-S06.py
question09-S06.py
question02-S06.py
question05-S06.py
#question05-S06
students = students = {
'Ali': [18, 17, 20],
'Sara': [15, 19, 18],
'Reza': [12, 14, 10],
'Mina': [20, 20, 19] }
for i in students:
avg = sum(students[i])/len(students[i])
print(f"{i},{avg: .2f}")
a = max(students[i])
print(f'highest score:{a}')
if avg >=15 :
print('Status: Passed\n')
else:
print("Status: Failed\n")
best = max({avg})
print(f"The best average: {i}-->{best:.2f}")
Name Type Size Value
a int 1 20
average float 1 3433.3333333333335
Help Variable Explorer Debugger Profiler Plots Files
Console 1/A
In [49]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/Session-06/question05-S06.py' --wdir
Ali, 18.33
highest score:20
Status: Passed
Sara, 17.33
highest score:19
Status: Passed
Reza, 12.00
highest score:14
Status: Failed
Mina, 19.67
highest score:20
Status: Passed
The best average: Mina-->19.67
In [50]:
IPython Console History
Inline Conda: anaconda3 (Python 3.13.9) LSP: Python Line 22, Col 42 UTF-8 CRLF RW Mem 78%


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question06-S06.py
question01-S06.py
question03-S06.py
question04-S06.py
question06-S06.py
question08-S06.py
1 sales = (('Ali', 'Laptop', 1200),('Sara', 'Phone', 800),('Ali', 'Phone', 800),\
2 ('Reza', 'Laptop', 1200),('Sara', 'Laptop', 1200),('Ali', 'Mouse', 50))
3
4 names = set(i[0] for i in sales)
5
6 for name in names:
7 total=[]
8 s = sum([i[2] for i in sales if i[0]==name])
9 total.append((name,s))
10 print(f'Customer perchase: {total}\n')
11
12 for i in total:
13 x = max([i[1]]for i in total)
14 print(f'Max purchase: {name,x}\n')
15
16
17 b = [i[1]for i in sales]
18 for i in b:
19 if b.count(i) >1:
20 print(f'Best selling products: {i}\n')
21 break
22
23 c = [i[1] for i in sales]
24 for i in set(c):
25 print(f'product sales:{i}-->{c.count(i)}\n')
26
27 total_sales = [i[2] for i in sales]
28 total_sales = sum(total_sales)
29 print(f'Total sales: {total_sales} $')
Name
Date Modified
Q4-s06.png
16-Sep-26 12:12 PM
Console 1/A
In [113]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py
Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/
Session-06/question06-S06.py' --wdir
Customer perchase: [('Reza', 1200)]
Customer perchase: [('Sara', 2000)]
Customer perchase: [('Ali', 2050)]
Max purchase: ('Ali', [2050])
Best selling products: Laptop
product sales:Phone-->2
product sales:Laptop-->3
product sales:Mouse-->1
Total sales: 5250 $
In [114]:
IPython Console
History


File Edit Search Source Run Debug Consoles Projects Tools Window Help
...ouz\python-exercises-sogol-karamouz\Session-06
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question07-S06.py
question01-S06.py
question03-S06.py
question04-S06.py
question06-S06.py
question07-S06.py
#question07-S06
orders = [('Ali', 'Laptop'),('Sara', 'Phone'),('Ali', 'Phone'),\
('Reza', 'Laptop'),('Sara', 'Laptop'),('Ali', 'Tablet'),\
('Reza', 'Phone')]
dic = {}
for i in orders :
if i[0] not in dic :
dic[i[0]]=[]
dic[i[0]].append(i[1])
print(dic)
Name
Date Modified
Q4-s06.png
16-Sep-26 12:12 PM
Console 1/A
In [65]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/Session-06/question07-S06.py' --wdir
{'Ali': ['Laptop', 'Phone', 'Tablet'], 'Sara': ['Phone', 'Laptop'], 'Reza': ['Laptop', 'Phone']}
In [66]:
IPython Console
History

Spyder (Python 3.13)

File Edit Search Source Run Debug Consoles Projects Tools Window Help

C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question08-S06.py

question01-S06.py × question03-S06.py × question04-S06.py × question06-S06.py × question08-S06.py ×

```python
#question08-S06
users = [
('Ali', 25, 'Python'),
('Sara', 30, 'Java'),
('Reza', 22, 'Python'),
('Mina', 28, 'C++'),
('John', 35, 'Python'),
('David', 30, 'Java')
]


d = {language:[i[0]for i in users if i[2]==language] for language in set(i[2] \
                                                    for i in users)}

print(d,'\n')

avg_age = sum( i[1] for i in users)/len(users)
print(f'average age is : {avg_age:.2f}\n')

oldest = max(i[1] for i in users)
print(f'The oldest person is : {oldest} y/o\n')

most_user= max(d.keys())

print(f'Language with most users: {most_user}\n')

print(f'Languages: {set(d.keys())}')
```

...ouz\python-exercises-sogol-karamouz\Session-06

Name	Date Modified
Q4-s06.png	16-Sep-26 12:12 PM

Console 1/A ×

```
In [115]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py
Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/
Session-06/question08-S06.py' --wdir
{'C++': ['Mina'], 'Java': ['Sara', 'David'], 'Python': ['Ali', 'Reza', 'John']}

average age is : 28.33

The oldest person is : 35 y/o

Language with most users: Python

languages: {'Python', 'Java', 'C++'}

In [116]:
```

IPython Console History

Inline Conda: anaconda3 (Python 3.13.9) LSP: Python Line 15, Col 10 UTF-8 CRLF RW Mem 74%

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Spyder (Python 3.13)

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C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-06\question09-S06.py

uestion06-S06.py × question08-S06.py × question09-S06.py × question02-S06.py × question05-S06.py ×

```python
#question09-S06

products = {
'P01': ('Laptop', 1200, 5),
'P02': ('Phone', 800, 0),
'P03': ('Tablet', 500, 12),
'P04': ('Mouse', 50, 25),
'P05': ('Keyboard', 100, 0)
}

product_name = [i[0] for i in products.values()]
print(f'Products_name: {product_name}')

Available = [(i[0],i[2]) for i in products.values() if i[2]>0]
unavailable = [(i[0],i[2]) for i in products.values() if i[2]==0]
d = dict(Available)
d1=dict(unavailable)
print(f'Available products: {dict(d)}')
print(f'Unavailable products: {dict(d1)}')

Total = [(i[0],i[1]*i[2]) for i in products.values()]
for i in Total:
    print(f'Total stock values: {i[0],i[1]}')

print(f'Max Stock value : {max(Total)}')

tsv = sum([(i[1]*i[2]) for i in products.values()])

print(f'Total stock value: {tsv}')
```

...ouz\python-exercises-sogol-karamouz\Session-06

Name	Date Modified
Q4-s06.png	16-Sep-26 12:12 PM

Console 1/A ×

```
In [117]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py
Fanavari/python-exercises-sogol-karamouz/python-exercises-sogol-karamouz/
Session-06/question09-S06.py' --wdir
Products_name: ['Laptop', 'Phone', 'Tablet', 'Mouse', 'Keyboard']
Available products: {'Laptop': 5, 'Tablet': 12, 'Mouse': 25}
Unavailable products: {'Phone': 0, 'Keyboard': 0}
Total stock values: ('Laptop', 6000)
Total stock values: ('Phone', 0)
Total stock values: ('Tablet', 6000)
Total stock values: ('Mouse', 1250)
Total stock values: ('Keyboard', 0)
Max Stock value : ('Tablet', 6000)
Total stock value: 13250

In [118]:
```

IPython Console History

Inline Conda: anaconda3 (Python 3.13.9) ✓ LSP: Python Line 6, Col 1 UTF-8 CRLF RW Mem 74%

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